

Beat Tracking

AI in Music, Winter Term 2025/2026

15.01.2026

Exercise 1 [Peak Picking, 5 points]

Read through the FMP notebook on peak picking and **run the code on your own device**. What is **missing from the notebooks is a method to quantitatively evaluate the peak picking methods** past the visualisations. Research and choose a metric for evaluating the accuracy of the picked peaks given the **annotated beats**, and use this metric to recommend a peak picking method for the example song given in the notebook.

Exercise 2 [Autocorrelation Tempogram, 5 points]

Implement the approach for the creation of tempograms given on slides 38 to 40 of the “Rhythm Analysis” lecture slides. Use the same audio from Exercise 1 to test your implementation. Aside from the global maximum in the tempogram, indicating the most likely tempo at any given time, there are other local maxima spread across the tempogram. Explain why these distinct other maxima are created, and how they relate to the true tempo.